

HKL ICA Matrix in the Bolt Pipeline

Fact-checked against the current trainer and affine-node code.

Short answer. Yes, conceptually:

PCA $\rightarrow$ ICA $\rightarrow$ component mask $\rightarrow$ inverse ICA $\rightarrow$ PCA back-projection.

The “identity matrix to remove components” is a diagonal mask: artifact ICs get 0, kept ICs get 1.

$D = \text{diag}(d)$, $d_i = 0$ for bad ICs, $d_i = 1$ otherwise `D = np.eye(k); D[bad,bad] = 0`

Column-vector form, full-rank ICA. For one channel vector $x \in \mathbb{R}^n$:

$$x_{\text{out}} = Cx, \quad C = P^\top ADWP.$$

$$\mathbf{x_out} = \mathbf{C} @ \mathbf{x} \quad \mathbf{C} = \mathbf{P.T} @ \mathbf{A} @ \mathbf{D} @ \mathbf{W} @ \mathbf{P}$$

P retained PCA projection
 W ICA unmixing matrix
 D diagonal keep/remove mask for ICA components
 A ICA mixing matrix, i.e. inverse ICA in the retained space
 $P^\top$ PCA back-projection

Bolt row-vector form. Bolt applies row samples by right multiplication:

$$X_{\text{out}} = XM, \quad M = C^\top = (DWP)^\top (P^\top AD)^\top.$$

$$\mathbf{X_out} = \mathbf{X} @ \mathbf{M} \quad \mathbf{M} = \mathbf{C.T} \quad \mathbf{M} = \mathbf{unmixing.T} @ \mathbf{mixing.T}$$

The trained-channel block is embedded into a full-stream affine matrix; untrained channels are identity.